

Let  $p := \lfloor \theta n \rfloor$ . Take  $4p$  mutually disjoint vertex sets  $S_1, S_2, \dots, S_{2p}, T_1, T_2, \dots, T_{2p}$  from  $V(H) \setminus V(P')$  with  $i_P(S_i) = (m, k - \ell - m)$  and  $i_P(T_i) = (m - 1, k - \ell - m + 1)$  for  $i \in [2p]$ . We denote the union of these  $S_i$  and  $T_i$ ,  $i \in [2p]$  by  $R_1$ . So  $R_1 \subseteq V(H) \setminus V(P')$ ,  $|R_1| = 4(k - \ell)p$  and  $i_P(R_1) = 2p(m, k - \ell - m) + 2p(m - 1, k - \ell - m + 1)$ . Thus  $P'$  can absorb  $R_1$ , that is, there exists an  $\ell$ -path  $P$  with the same ordered ends as  $P'$ , where  $V(P) = V(P') \cup R_1$ . Now we show that  $P$  is the desired absorbing  $\ell$ -path. Note that  $|V(P)| \leq \gamma n$ . Fix any set  $X \subseteq V(H) \setminus V(P)$  with  $|X| \leq p$  and  $(k - \ell) \mid |X|$  as required by the lemma. Suppose further that  $i_P(X) = (t, s) = (x(m, k - \ell - m) + y(m - 1, k - \ell - m + 1))$ . Then we have

$$\begin{cases} x = t - \frac{(m-1)(t+s)}{k-\ell} \\ y = \frac{(t+s)m}{k-\ell} - t. \end{cases}$$

Since  $t+s = |X|$ ,  $(k-\ell) \mid |X|$  and  $m/(k-\ell) \leq 1$ , we get that  $x, y$  are integers and  $|x|, |y| \leq |X| < 2p$ . Thus  $i_P(X \cup R_1) = (x+2p)(m, k-\ell-m) + (y+2p)(m-1, k-\ell-m+1)$ , where  $x+2p > 0, y+2p > 0$  and  $|X \cup R_1| \leq 4(k-\ell)p + p \leq (k-\ell)\beta^{2k-2\ell+4}n$ . So  $P'$  can absorb  $X \cup R_1$ , that is,  $P$  can absorb  $X$ .  $\square$

To complete the proof of Lemma 3.5, it remains to prove Claim 3.9.

*Proof of Claim 3.9.* Let  $S := \{v_1, v_2, \dots, v_{k-\ell}\}$  be a  $(k-\ell)$ -set. Fix a  $k$ -partite  $k$ -graph  $A(k, \ell)$  on  $[V_1^A, V_2^A, \dots, V_k^A]$  satisfying Proposition 3.7 with  $|A(k, \ell)| = r \leq k^4$  and  $V(A(k, \ell)) = S' \cup X$ , where  $|S'| = k - \ell$ . Without loss of generality, suppose  $|S' \cap V_i^A| = 1$  for  $i \in [k - \ell]$ . Let  $b := (4k - 2\ell - 1)(k - \ell) + r$ . Since  $(a, k - a) \in I_P^\mu(H')$ , the number of edges whose index vectors are  $(a, k - a)$  is at least  $\mu n^k$ . Note that  $t_1 := |S \cap V_1| \leq a$  and  $|S \cap V_2| \leq k - a$  by  $S \in \mathcal{S}$ . Since  $A(k, \ell)$  is  $k$ -partite, by the supersaturation result (see [6]) on the subgraph of  $H$  that consists of all edges of index vector  $(a, k - a)$ ,  $H$  contains  $\beta n^r$  copies of  $A(k, \ell)$  each with  $V_1^A, \dots, V_{t_1}^A \subseteq V_1$  and  $V_{t_1+1}^A, \dots, V_{k-\ell}^A \subseteq V_2$ . For such a copy  $A$  of  $A(k, \ell)$ , we denote by  $S'_A$  as the set of  $k - \ell$  vertices given in Proposition 3.7 (2). So  $i_P(S'_A) = i_P(S)$  for each such  $A$ .

Consider a copy of  $A(k, \ell)$  in  $H$  which we denote by  $A$ . Note that each of  $V_1, V_2$  is  $(\beta, 2)$ -closed in  $H$ . Without loss of generality, suppose  $S'_A = \{w_1, w_2, \dots, w_{k-\ell}\}$  such that  $v_i, w_i$  are  $(\beta, 2)$ -reachable for  $i \in [k - \ell]$  by  $i_P(S'_A) = i_P(S)$ . By the definition of reachability, for each  $i \in [k - \ell]$ , there are at least  $\beta n^{4k-2\ell-1}$   $(4k - 2\ell - 1)$ -sets  $T_i$  such that there exist  $\ell$ -paths  $P_i^1, P_i^2, P_i^3, P_i^4$  with  $V(P_i^1 \cup P_i^2) = T_i \cup \{v_i\}$  and  $V(P_i^3 \cup P_i^4) = T_i \cup \{w_i\}$ , where  $P_i^1$  has the same ends as  $P_i^2$ , and  $P_i^3$  has the same ends as  $P_i^4$ . So there are at least  $\beta^{k-\ell+1} n^b$  choices for  $A \cup T_1 \cup T_2 \cup \dots \cup T_{k-\ell}$  as an ordered set. Among them, at most  $(k - \ell)n^{b-1}$  of them intersect  $S$  and at most  $b^2 n^{b-1}$  of them contain repeated vertices. Thus there are at least  $\beta^{k-\ell+1} n^b / 2$   $b$ -tuples avoiding  $S$  such that  $A, T_1, T_2, \dots, T_{k-\ell}$  are pairwise vertex-disjoint.

Now it remains to show that the  $b$ -tuple corresponding to  $A \cup T_1 \cup T_2 \cup \dots \cup T_{k-\ell}$  is an  $S$ -absorber. Firstly,  $T_i \cup \{w_i\}$ ,  $i \in [k - \ell]$  spans two vertex-disjoint  $\ell$ -paths of length two, which together with the spanning  $\ell$ -path in  $A \setminus \{w_1, w_2, \dots, w_{k-\ell}\}$  form a family of  $2k - 2\ell + 1$   $\ell$ -paths which span  $V(A) \cup T_1 \cup T_2 \cup \dots \cup T_{k-\ell}$ . Secondly,  $H[T_i \cup \{v_i\}]$  forms two vertex-disjoint  $\ell$ -paths of length two for  $i \in [k - \ell]$ , which together with the spanning  $\ell$ -path in  $A$  gives a family of  $2k - 2\ell + 1$   $\ell$ -paths which span  $S \cup V(A) \cup T_1 \cup T_2 \cup \dots \cup T_{k-\ell}$  and have the same ends as the family of  $\ell$ -paths above. So the  $b$ -tuple corresponding to  $A \cup T_1 \cup T_2 \cup \dots \cup T_{k-\ell}$  is an  $S$ -absorber (cf. Figure 1).  $\square$

**3.3. Proofs of Theorem 1.4 and Theorem 1.5.** We prove Theorem 1.4 by following the common approach of absorption (cf. [18, 27]). That is, we decompose the proof in the usual way into the absorbing path lemma, the reservoir lemma, the connecting lemma and the path cover lemma. We will use these lemmas to find an absorbing path and a reservoir firstly, and then we cover the majority of vertices by vertex-disjoint  $\ell$ -paths. We connect up all these  $\ell$ -paths to form an  $\ell$ -cycle. Finally, we absorb the leftover vertices into the absorbing path, thereby completing a Hamilton  $\ell$ -cycle.